

Assignment 09

Fundamental Data Types

Due before class on June 24 (Tuesday)

Complete these three problems at the end of Chapter 4: P4.4, P4.6, and P4.12. I've modified the instructions just a little from your textbook.

For Exercise P4.4, you don't need to implement the portion of the program that asks the user for two numbers and then prints the output. Instead, implement the **Pair** class, with accessor methods named **getSum()**, **getDifference()**, **getProduct()**, **getAverage()**, **getDistance()**, **getMaximum()** and **getMinimum()**, along with the necessary constructor. Include a JUnit test class, **PairTest** (not **Tester**) that tests each of the methods.

For Exercise P4.6 you'll write the class **DataSet** that includes a constructor as well as the mutator, and two accessors specified in the problem. The **DataSetTest** (not **Tester**) program should be a JUnit test program that tests each of the methods, not a program that prints out the actual and expected values.

Like the other problems, for Exercise P4.12 you won't need to write the interactive program that prompts the user for an integer and that prints out the various values. Instead, simply write the **DigitExtractor** class as well as a JUnit class **DigitExtractorTest** that tests **nextDigit()** repeatedly. To get the first digit (that is, the 10's place) use the % 10 operator. To get the next digit, divide the original number by 10 to move each digit one position to the right.

Submit the entire project to Web-Cat before the deadline. You can use the BlueJ Submitter plugin, (like we did in class), or you can just upload the files through the Web interface. To upload the files using your browser, you'll need to zip up both of the files manually; it doesn't matter what you name the zip file. You can submit the homework assignment three times.

This assignment will be graded on a combination of the percentage of passing tests, the portion of your class actually tested by your own tests (called the test coverage percentage), a set of tests that I've designed, and 10% on style. Remember that the Web-Cat scoring is set to 50 points, which will be scaled down to 5 points.